AGENTIC AI

Date:

ASSIGNMENT ALPHA

Q1:

1) $X = \frac{29491200}{1280 \times 720} = 32$ bits/pixel

2) $\frac{29491200}{8} = 3686400$ bytes/frame

3) $(70 \times 10^6) \times 8 = 560\,000\,000$ bits/s

Q2:

$$\Rightarrow Q = \begin{bmatrix} 1.5 & 3 \\ 2 & 4 \\ 2.2 & 4.4 \\ 2.4 & 4.4 \end{bmatrix}, \quad K = \begin{bmatrix} 1.1 & 2.2 \\ 1 & 3 \\ 1 & 3 \\ 1.1 & 3 \end{bmatrix}, \quad V = \begin{bmatrix} 0.2 & 0.3 \\ 2 & 0.3 \\ 2.2 & 0.3 \\ 2.2 & 0.3 \end{bmatrix}$$

$$\Rightarrow QK^T = \begin{bmatrix} 8.25 & 10.5 & 10.5 & 10.65 \\ 11 & 14 & 14 & 14.2 \\ 12.1 & 15.4 & 15.4 & 15.62 \\ 13.32 & 15.6 & 15.6 & 15.84 \end{bmatrix}$$

$$\Rightarrow \frac{QK^T}{\sqrt{2}} = \begin{bmatrix} 5.834 & 7.425 & 7.425 & 7.531 \\ 7.778 & 9.899 & 9.899 & 10.041 \\ 8.556 & 10.889 & 10.889 & 11.045 \\ 8.712 & 11.031 & 11.031 & 11.201 \end{bmatrix}$$

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$$\Rightarrow \text{softmax}\left(\frac{QK^T}{\sqrt{2}}\right) = \begin{bmatrix} 0.0614 & 0.3016 & 0.3016 & 0.3354 \\ 0.0366 & 0.3056 & 0.3056 & 0.3521 \\ 0.0297 & 0.3063 & 0.3063 & 0.3578 \\ 0.0300 & 0.3046 & 0.3046 & 0.3609 \end{bmatrix}$$

⇒ Multiply by V:

$$\text{Attention Scores} = \begin{bmatrix} 2.0168 & 0.3000 \\ 2.0636 & 0.3000 \\ 2.0794 & 0.3000 \\ 2.0792 & 0.3000 \end{bmatrix}$$

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→ Governance
→ Emerging
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Q3:

⇒ Given training data:

Wind	Humidity	Rain
Weak	High	Yes
Strong	High	Yes
Weak	Low	No
Weak	low	Yes
Strong	low	No

⇒ Total = 5
⇒ Rain (Yes) = 3
Rain (No) = 2

1.

⇒ P(Yes) = 3/5 = 0.6
P(No) = 0.4

⇒ Laplace Smoothing:

$$P(x|C) = \frac{count(x,C) + 1}{count(C) + k}$$

where k = 2, because each feature has two possible values.

⇒ For Rain = Yes:

↳ For wind:

$$P(weak|Yes) = \frac{2+1}{3+2} = 0.6$$

$$P(strong|Yes) = 0.4$$

↳ For Humidity:

$$P(high|Yes) = \frac{2+1}{3+2} = 0.6$$

$$P(low|Yes) = 0.4$$

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⇒ For Rain = No:

↳ For wind.

$$P(\text{weak} | \text{No}) = \frac{1+1}{2+2} = 0.5$$

$$P(\text{strong} | \text{No}) = \frac{1+1}{2+2} = 0.5$$

↳ For Humidity:

$$P(\text{high} | \text{No}) = \frac{0+1}{2+2} = 0.25$$

$$P(\text{low} | \text{No}) = \frac{2+1}{2+2} = 0.75$$

⇒ Case 1: Weak wind, High humidity

↳ For yes:

$$P(\text{Yes}) \times P(\text{weak} | \text{Yes}) \times P(\text{high} | \text{Yes})$$

$$0.6 \times 0.6 \times 0.6$$

$$0.216$$

↳ For No:

$$0.4 \times 0.5 \times 0.25$$

$$0.05$$

↳ Normalized Probability:

$$P(\text{Yes} | \text{weak, high}) = \frac{0.216}{0.216 + 0.5} = 0.812$$

Hence, Rain = 4, drone should not deliver.

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⇒ Case 2: Strong wind, High humidity

↳ P(Yes) = 0.6 × 0.4 × 0.6 = 0.144

↳ P(No) = 0.4 × 0.5 × 0.25 = 0.05

↳ Normalized:

$$\frac{0.144}{0.144 + 0.05} = 0.742$$

hence, Rain = Yes.

⇒ Case 3: Weak wind, low Humidity.

↳ P(Yes) = 0.6 × 0.6 × 0.4 = 0.144

↳ P(No) = 0.4 × 0.5 × 0.75 = 0.150

↳ 0.150 > 0.144

hence, Rain = No.

⇒ Case 4: Strong wind, low Humidity

↳ P(Yes) = 0.6 × 0.4 × 0.4 = 0.096

↳ P(No) = 0.4 × 0.5 × 0.75 = 0.150

↳ Hence, Rain = No.

⇒ Training Accuracy = 4/5 × 100 = 80%.

Date:

2. Entropy and Information Gain:

$\Rightarrow H(S) = -\frac{3}{5}\log_2\frac{3}{5} - \frac{2}{5}\log_2\frac{2}{5} = 0.971$

$\Rightarrow$ IG of wind:

↳ for weak wind:

$H(\text{weak}) = -\frac{2}{3}\log\frac{2}{3} - \frac{1}{3}\log\frac{1}{3} = 0.918$

↳ for strong wind:

$H(\text{strong}) = 1$

↳ Weighted Entropy:

$H(S|\text{wind}) = 3/5(0.918) + 2/5(1) = 0.951$

↳ $IG(\text{wind}) = H(S) - H(S|\text{wind})$

$= 0.020$

$\Rightarrow$ IG of Humidity:

↳ $H(\text{low}) = -\frac{1}{3}\log(\frac{1}{3}) - \frac{2}{3}\log\frac{2}{3} = 0.918$

↳ $H(\text{high}) = 0$

↳ $H(S|\text{Humidity}) = \frac{2}{5}(0) + \frac{3}{5}(0.918) = 0.551$

↳ $IG(\text{Humidity}) = * 0.971 - 0.551$

$= 0.420$

$\Rightarrow$ IG(Humidity) > IG(wind) therefore Humidity is selected as the root node.

Date:

→ Low-Humidity Branch:

↳ Weak, Low → No

Weak, low → Yes

Strong, low → No

↳ Entropy:

$$H(low) = 0.918$$

↳ Splitting on wind:

↳ for strong:

$$H(strong) = 0$$

↳ for weak =

$$H(weak) = 1$$

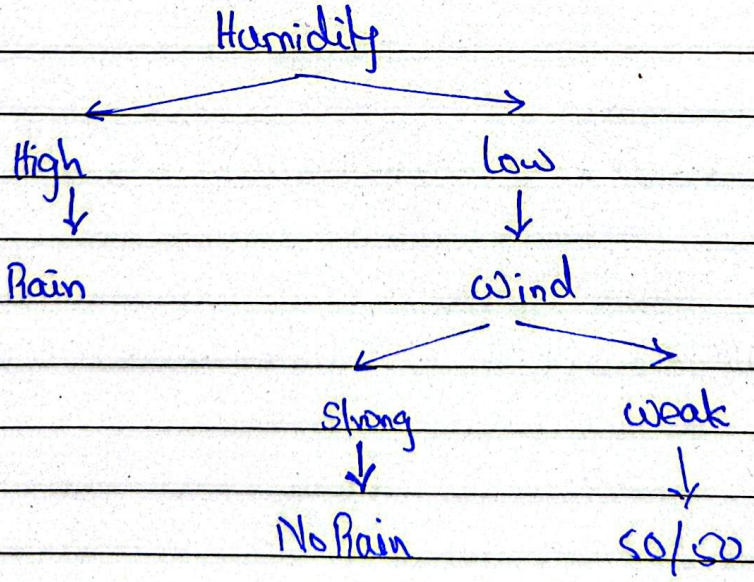
↳ weighted entropy:

$$\frac{2}{3}(1) + \frac{1}{3}(0) = 0.667$$

↳ $IG(Wind \mid low) = 0.918 - 0.667$

$= 0.252$

↳ Training Accuracy = 80%.



Date:

Q5.

1) $P(A = \text{Infected}) = 0.5$

$P(B = \text{Infected}) = 0.5$

$P(A=I, B=I) = 0.5 \times 0.5$

$= 0.25$

2)

⇒ Initial State: $(A=I, B=I)$

Drone starts at A

⇒ $t=0$:

↳ A = Infected

↳ Action = Spray A

↳ Credit = $0

↳ Resultant State: $(A=H, B=I)$

⇒ $t=1$:

↳ Drone perceives A again and finds it to be healthy.

↳ Credit = $2

↳ Action = Move to B

↳ Resultant State: $(A=H, B=I)$

⇒ $t=2$

↳ B = Infected

↳ Action = Spray B

↳ Credit = $0

↳ Resultant State = $(A=H, B=H)$

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⇒ t=3

↳ B = Healthy

↳ credit = $2

↳ Action = Move to A

↳ Resultant State : (A=H, B=H)

⇒ Final Table:

Time	Percept	Action	Credit	Resultant State
0	[A=I]	SprayA	0	A=H, B=I
1	[A=H]	Move to B	2	A=H, B=I
2	[B=I]	Spray B	0	A=H, B=H
3	[B=H]	Move to A	2	A=H, B=H

⇒ Total Credit = $4

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Q6:

→ PAPER 1:

1) Game theory models agent interactions as a game including players, actions, strategies, and payoffs, it gives a baseline of rational behavior. We can then compare what LLM agents actually do against that baseline, and the gap is measured as bias.

2) Game theory explains cooperation, competition, and coordination. As AI agents negotiate and decide for the users, it gives a principled way to check whether they behave predictably and fairly

3) The two game theories presented in the paper are Prisoner's Dilemma and Battle of the Sexes. According to Prisoner's Dilemma, each player's dominant strategy is to defect, so both defect even though mutual cooperation pays more. Meanwhile, according to the Battle of the Sexes, players prefer different outcomes but both prefer coordinating to disagreeing, so there are two equilibria. These can be used for the 5G applications as the Prisoner's Dilemma fits power control and spectrum sharing; transmitting at full power helps but raises interference for everyone. Battle of the Sexes fits channel or band selection, network-slice coordination and edge-server selection, where operators must agree on one option but each prefers its own.

Date:

4) Llama 3.1 405B (Meta)
Mistral large
GPT 4
Claude 3.5 Sonnet.

⇒ PAPER 2:

1) The paper models the problem as a Dec-POMDP:
$G = (N, S, A, P, R, \Omega, O, n, \gamma)$
where, 'N' is the set of agents, 'S' the global states, 'A' the joint actions, and $P(s'|s,a)$ the transition function.
'R' is the reward function, 'Ω' the observation set, and 'O' the observation function. Each agent sees only a local observation.
'n' is the number of agents, and 'γ' is the discount factor.

2) Hard Parameter Sharing:
Agents share common layers and keep separate output heads. It is efficient and works when tasks have common structure.
Soft Parameter Sharing:
Each agent keeps its own model, and the parameters are only pushed to stay similar. It allows more specialization.
Applications:
Multitask learning, heterogeneous MARL, and deploying several models on constrained devices.

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3) $L = E_D \sum_i \left[\| f_o(o_i, a_i, z_i) - o'_i \|^2 + \lambda_e (f_r(o_i, a_i, z_i) - r_i)^2 \right]$

where:

o_i : current observation

a_i : action

z_i : latent representation

f_o : next observation predictor

f_r : reward predictor

o'_i : next observation

r_i : actual reward

λ_i : weighting parameter.

4) It shows how the training signal is built from the environment reward plus two intrinsic rewards:

i) Exploration Reward:

Higher when the state is rarely seen. It pushes agents to explore early.

ii) Diversity Reward:

Keeps the representations of the same agent type close and different types apart, so shared layer dont blur different roles together.

Date:

Q7

1)

⇒ $A = (6, -1, \sqrt{9}) = (6, -1, 3)$

⇒ $b = (4, X, -2)$

⇒ $A \cdot b = 0$

$(6)(4) + (-1)(X) + (3)(-2) = 0$

$18 - X = 0$

$X = 18$

2)

⇒ Dot-Product:

$a \cdot b = a_1b_1 + a_2b_2 + a_3b_3$

here the output is a scalar

⇒ Hadamard Product:

$a \odot b = (a_1b_1, a_2b_2, a_3b_3)$

here the output is a vector.

3)

⇒ $u = (1, 2)$

⇒ $v = (2, 2)$

⇒ L2 Norms:

$\|u\| = \sqrt{1^2 + 2^2} = \sqrt{5}$

$\|v\| = \sqrt{2^2 + 2^2} = \sqrt{8}$

⇒ $\theta_v = 45°$

$\theta_u = \tan^{-1}(2/1) = 63.435°$

⇒ Angle between them $= \theta = 18.435°$

⇒ $\cos\theta = 3/\sqrt{10}$

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⇒ Projection of u onto the direction of v:

$$\|u_{proj}\| = \|u\| \cos\theta$$

$$= \frac{3}{\sqrt{2}}$$

⇒ $u \cdot v = \|u_{proj}\| \|v\|$

$$= \frac{3}{\sqrt{2}} (2\sqrt{2})$$

$$= 6$$

Q8:

⇒ $S(t) = 0.2t + t^2 + 2t^5 + t^6 + t^8 + 3\sin t + 2\cos t$

⇒ $t = 1°$:

↳ Agent A:

$A(1) = 0.2(1) + 1^2 = 1.2$

↳ Agent B:

$B(1) = \sin(1) + \cos(1) = 1.0173$

↳ Agent C:

$C(1) = 1^5 + 1^6 + \sin(1) = 2.0175$

↳ Agent D:

$D(1) = 1^5 + 1^8 + \sin(1) + \cos(1) = 3.0173$

↳ $S(1) = 1.2 + 1.0173 + 2.0175 + 3.0173 = 7.2521.$

⇒ $t = 2°$:

↳ A(2) = 4.4

↳ B(2) = 1.0343

↳ C(2) = 96.0349

↳ D(2) = 289.0343

↳ S(2) = 390.5035.

⇒ $t = 3°$:

↳ A(3) = 9.6

↳ B(3) = 1.0510

↳ C(3) = 972.0523

↳ D(3) = 6805.0510

↳ S(3) = 7737.7543

⇒ $t = 4°$:

↳ A(4) = 16.8

↳ B(4) = 1.0673

↳ C(4) = 5120.0698

↳ D(4) = 66561.0673

↳ S(4) = 71699.0044

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⇒ The distribution is polynomial as the term t^a dominates as time passes.

